

Daniel Yu

 Portfolio

 GitHub

 LinkedIn

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 512-968-5454

EDUCATION

MS, Electronic Systems Engineering

Texas A&M University

Graduation: May 2026

BS, Electronic Systems Engineering

Texas A&M University

*Minors: **Computer Science & Cybersecurity***

Graduation: May 2024

WORK EXPERIENCE

Graduate Researcher

College Station, TX

ETID Department at TAMU

Aug 2024 - Present

- Developed NDT techniques and Digital Twins for building materials, focusing on reinforced concrete structures.
- Utilized microwave and SAR imaging algorithms with Machine Learning for analysis of structural components.
- Implemented embedded systems, analog electronics, and RF solutions for material evaluation methodologies.

Graduate Teaching Assistant

College Station, TX

ETID Department at TAMU

Aug 2024 – Present

- Instructed 180+ undergraduates circuit analysis and design using semiconductor devices such as diodes, transistors, and op-amps for applications like power supplies, filters, amplifiers, and signal conditioning.
- Used Multisim for schematic simulation and pre-lab validation of voltage/current behavior in electronic systems.
- Led labs in testing and troubleshooting using oscilloscopes, multimeters, function generators to validate performance.

Automation Engineer Intern

Beijing, China

HangZhen Technology Co., Ltd

May 2023 – July 2023

- Programmed Siemens PLCs to control machine tool sequencing and sync operations with PCS for better workflow.
- Reviewed and documented PLC programs for multi-station automation systems to support integration and verification.
- Contributed to the standardization of project execution workflows.

RELEVANT EXPERIENCE

Spaceshooter Game

- Implemented a real-time space shooter in C++ using the Raylib library, featuring dynamic player/enemy behavior and input-based projectile systems.
- Built OOP architecture for interactive entities (player, enemies, projectiles), including collision detection and event-driven bullet spawning.
- Developed game loop, physics logic, and real-time input handling; integrated AI-generated sprite assets for visuals.

Time Tracking Software

- Developed a desktop time tracking application in Python using Tkinter, featuring a GUI for structured task input, category assignment, and real-time session tracking
- Implemented a project-task hierarchy system to simulate real-world productivity workflows, enabling nested grouping of items under broader timelines
- Integrated event-driven logging that captures task data upon completion and writes structured entries to a persistent file for future reference

HRTBT Group - Embedded Software Engineer

- Implemented data buffering, synchronization, and preprocessing pipelines in embedded C to support FFT-based range and Doppler extraction.
- Applied embedded API wrappers to bridge between proprietary firmware drivers and custom signal processing routines for real-time execution.
- Configured TI mmWave radar sensors via TI SDK, modifying config files and parsing UART output for raw ADC data acquisition.

Text Adventure Game

- Developed a branching text adventure game in ARM Cortex-M assembly on the MSP432, using conditional logic and jump instructions to manage narrative state transitions.
- Programmed GPIO-based input handling and LCD control to display story content dynamically based on player input and narrative state.

Path Finding Robot

- Integrated IR sensing, digital control logic, and motor actuation in a real-time closed-loop system on an FPGA board.
- Implemented a VHDL-based FSM to control motor speed and direction using real-time IR sensor input.
- Generated PWM signals to control dual H-bridge drivers for smooth motion and path correction.

Muscle-Driven Control Logic System

- Designed an analog front-end with calibrated amplification and filtering to extract clean EMG signals.
- Implemented threshold-based control logic in LabVIEW to convert EMG inputs into real-time directional commands.
- Integrated data acquisition and microcontroller interfacing into a closed-loop control system via UART communication.

PID Control for Inverted Pendulum

- Built an inverted pendulum system with DC motor control and encoder-based position feedback.
- Implemented a discrete PID controller on Arduino DUE to stabilize pendulum angle based on position error.
- Applied high-frequency sampling and PWM to maintain pendulum equilibrium under disturbance.

SKILLS

Programming Languages & Software Tools: C, C++, Python(Pandas, Numpy), MATLAB, VHDL, Git, LabVIEW, Java, JavaScript, HTML/CSS, Haskell, Assembly(ARM & x86)

Embedded Systems & Firmware Development: I2C, SPI, UART, FreeRTOS, TI-RTOS, Logic Analyzer & Oscilloscope Debugging, Arduino, Intel Quartus Prime

Electronics & Instrumentation: Analog & Digital Circuit Design, Power Electronics, Multisim, Altium Designer, Oscilloscope, Spectrum Analyzer, VNA, Semiconductor Test & Validation, ATE

Control Systems & Automation: Structured Text, Ladder Logic, Function Block Diagram, Sequence Function Chart, PIDs, Siemens Simatics, SCADA System, HMI Design, LabVIEW

Machine Learning & Data Science: Artificial Neural Networks(ANN), Machine Learning Algorithms, Scikit-learn, TensorFlow, PyTorch

RF & Electromagnetics: RF & Electromagnetics, Antenna Theory, Ansys HFSS, Signal Processing, Radar Imaging

Networking & Security: Cisco Networking Architecture, Network Systems/Security

HONORS

Dean's Honor Roll

Fall 2023, Spring 2024

Distinguished Student

Spring 2023